

3.4.2 Gravitation

- **Newton's law**

Gravity as a universal attractive force acting between all matter.

Force between point masses

$$F = \frac{Gm_1m_2}{r^2},$$

where G is the gravitational constant.

- **Gravitational field strength**

Concept of a force field as a region in which a body experiences a force.

Representation by gravitational field lines.

g as force per unit mass defined by $g = \frac{F}{m}$

Magnitude of g in a radial field given by $g = \frac{GM}{r^2}$

- **Gravitational potential**

Understanding of the definition of gravitational potential, including zero value at infinity, and of gravitational potential difference.

Work done in moving mass m given by

$$\Delta W = m \Delta V$$

Gravitational potential V in a radial field given by

$$V = -\frac{GM}{r}$$

Graphical representations of variations of g and V with r .

V related to g by $g = -\frac{\Delta V}{\Delta r}$

- **Orbits of planets and satellites**

Orbital period and speed related to radius of circular orbit.

Energy considerations for an orbiting satellite.

Significance of a geosynchronous orbit.